Chapter 4 - System Specification

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4.1 Introduction

This chapter presents the complete system specifications of the **Intelligent Task Allocation System (ITAS)**. It defines the functional and non-functional requirements that govern how the system operates and how it should perform under different conditions. These specifications translate the problem definition and objectives described in previous chapters into concrete, testable system requirements. They serve as a formal foundation for system design, implementation, testing, and evaluation.

The functional requirements describe *what* the system must do in terms of features, inputs, processing, and outputs. The non-functional requirements define *how well* the system must perform its functions, covering aspects such as performance, security, usability, reliability, and ethical AI considerations.

4.2 Functional Requirements

The functional requirements specify the core capabilities of ITAS. Each requirement is designed to be clear, testable, and directly traceable to the project objectives.

FR-01 User Registration

The system shall allow new users to register accounts as either **Project Managers** or **Employees**.

FR-02 User Authentication

The system shall allow registered users to securely log in and log out using valid credentials.

FR-03 Role-Based Access Control

The system shall enforce role-based permissions such that: - Project Managers can manage tasks, assignments, and reports. - Employees can manage profiles, upload CVs, view assigned tasks, and update task status.

FR-04 Employee Profile Management

The system shall allow employees to create and update personal profiles including name, job title, and professional information.

FR-05 CV and Portfolio Upload

The system shall allow employees to upload CV and portfolio documents in supported formats (e.g., PDF).

FR-06 Document Text Extraction

The system shall automatically extract text content from uploaded CV and portfolio documents.

FR-07 NLP-Based Skill Extraction

The system shall identify and extract technical skills, tools, and technologies from the extracted text using Natural Language Processing (NLP) techniques.

FR-08 Skill Repository Management

The system shall store extracted skills in a structured database and automatically update employee skill profiles.

FR-09 Task Creation

The system shall allow Project Managers to create tasks with attributes such as title, description, priority level, due date, and required skill tags.

FR-10 Task Modification and Deletion

The system shall allow Project Managers to update or delete existing tasks.

FR-11 Task Status Workflow

The system shall support task status transitions including *To Do*, *In Progress*, *Blocked*, *Completed*, and *Cancelled*.

FR-12 Workload Tracking

The system shall calculate and maintain each employee’s workload based on assigned and active tasks.

FR-13 Task Matching Request

The system shall allow Project Managers to request intelligent task-to-employee matching.

FR-14 Suitability Scoring

The system shall compute a suitability score for each employee–task pair based on: - Skill overlap between task requirements and employee skills - Current workload and availability

FR-15 Employee Ranking and Recommendation

The system shall generate a ranked list of recommended employees for each task along with suitability scores and matching skills.

FR-16 Task Assignment

The system shall allow Project Managers to assign tasks to employees based on system recommendations or manual selection.

FR-17 Assignment Notification

The system shall notify employees when tasks are assigned to them.

FR-18 Employee Task View

The system shall allow employees to view all tasks assigned to them, including deadlines and required skills.

FR-19 Task Progress Update

The system shall allow employees to update task status and provide progress notes.

FR-20 Management Dashboard

The system shall provide Project Managers with a dashboard displaying task progress, employee workload distribution, and overdue tasks.

FR-21 Report Generation

The system shall generate summary reports related to task allocation, workload distribution, and assignment history.

FR-22 Audit Logging

The system shall log critical system actions such as task creation, assignment, and status changes with timestam.

4.3 Non-Functional Requirements

4.3.1 Performance Requirements

- The system shall respond to user requests within acceptable time limits under normal load.
- The system shall efficiently process CV parsing and skill matching operations.

4.3.2 Security Requirements

- The system shall enforce secure authentication and authorization.
- User credentials and sensitive data shall be encrypted.
- Uploaded documents shall be protected from unauthorized access.

4.3.3 Usability Requirements

- The system shall provide a clear, intuitive web-based user interface.
- Dashboards shall present information visually using tables and charts.

4.3.4 Reliability Requirements

- The system shall handle errors gracefully without data loss.
- The system shall ensure data consistency during task assignment and updates.

4.3.5 Maintainability Requirements

- The system shall be modular and well-documented.
- The system shall allow future enhancements and integration with external systems.

4.3.6 Ethical and AI Considerations

- The system shall minimize bias in task allocation decisions.
- Suitability scoring shall be transparent and based on objective criteria.
- The system shall promote fairness by providing equal task opportunities based on verified technical skills.